The extension $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ as a nontrivial algebraic bundle

The exact sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ corresponds to the element $(1, 1, 1, \ldots) \in \prod_p \mathbb{Z}_p = \hat{\mathbb{Z}}$ under the isomorphism $\mathrm{Ext}^1(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) \cong \hat{\mathbb{Z}}$. This is the unit element in the profinite completion of the integers — the unique element whose projection to each p-adic component is the multiplicative identity. The extension decomposes prime-by-prime into the family of extensions $0 \to \mathbb{Z} \to \mathbb{Z}[1/p] \to \mathbb{Z}(p^\infty) \to 0$, each representing $1 \in \mathbb{Z}_p$, and these are assembled via pullback along the canonical decomposition $\mathbb{Q}/\mathbb{Z} \cong \bigoplus_p \mathbb{Z}(p^\infty)$. The fact that $\mathbb{Q}$ is the injective hull of $\mathbb{Z}$ (Wikipedia) endows this extension with a universal property: every extension of a torsion group by $\mathbb{Z}$ arises as a pullback of this one. Under Pontryagin duality, the sequence dualizes to the compact exact sequence $0 \to \hat{\mathbb{Z}} \to \mathbb{A}_{\mathbb{Q}}/\mathbb{Q} \to \mathbb{R}/\mathbb{Z} \to 0$, (nLab) where $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ is the adelic solenoid. (arXiv) (nLab)

Question 1: the extension class is $(1, 1, 1, \ldots) \in \prod_p \mathbb{Z}_p$

Computing $\mathrm{Ext}^1(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) \cong \hat{\mathbb{Z}}$

The computation proceeds in three steps. First, the primary decomposition $\mathbb{Q}/\mathbb{Z} \cong \bigoplus_p \mathbb{Z}(p^\infty)$ expresses the quotient as the direct sum of Prüfer p-groups over all primes. (jordanbell) Second, the standard identity for Ext applied to direct sums in the first argument gives

$$\mathrm{Ext}^1(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) \cong \mathrm{Ext}^1(\bigoplus_p \mathbb{Z}(p^\infty), \mathbb{Z}) \cong \prod_p \mathrm{Ext}^1(\mathbb{Z}(p^\infty), \mathbb{Z}).$$ (Wikipedia) (cambridge)

Third, each factor $\mathrm{Ext}^1(\mathbb{Z}(p^\infty), \mathbb{Z}) \cong \mathbb{Z}_p$ is computed by applying $\mathrm{Hom}(\mathbb{Z}(p^\infty), -)$ to the injective resolution $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$. (cambridge) Since $\mathrm{Hom}(\mathbb{Z}(p^\infty), \mathbb{Z}) = 0$ and $\mathrm{Hom}(\mathbb{Z}(p^\infty), \mathbb{Q}) = 0$ (a torsion group admits only the zero map to a torsion-free group) and $\mathrm{Ext}^1(\mathbb{Z}(p^\infty), \mathbb{Q}) = 0$ ($\mathbb{Q}$ is injective), the long exact sequence collapses to the isomorphism

$$\mathbf{Ext^1(\mathbb{Z}(p^\infty), \mathbb{Z}) \cong Hom(\mathbb{Z}(p^\infty), \mathbb{Q}/\mathbb{Z}) \cong End(\mathbb{Z}(p^\infty)) \cong \mathbb{Z}_p.}$$ (cambridge)

The last step uses the classical fact that any homomorphism $\mathbb{Z}(p^\infty) \to \mathbb{Q}/\mathbb{Z}$ lands in the p-primary component $\mathbb{Z}(p^\infty) \subset \mathbb{Q}/\mathbb{Z}$, and the endomorphism ring of the Prüfer p-group is the ring of p-adic integers (Fuchs, *Infinite Abelian Groups*, §44). An endomorphism is determined by the compatible sequence $(\varphi(1/p^n + \mathbb{Z}))_{n \geq 1}$, which defines an element of $\varprojlim_n \mathbb{Z}/p^n\mathbb{Z} = \mathbb{Z}_p$. (Encyclopedia of Mathematics) (Feog) Combining these factors: $\mathbf{Ext^1(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) \cong \prod_p \mathbb{Z}_p = \hat{\mathbb{Z}}}$. (Wikipedia)

Identifying the element

The extension class of $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ in $\prod_p \mathrm{Ext}^1(\mathbb{Z}(p^\infty), \mathbb{Z})$ is determined by its p-component for each prime p. The p-component is obtained by pulling back the extension along the inclusion $\iota_p: \mathbb{Z}(p^\infty) \hookrightarrow \mathbb{Q}/\mathbb{Z}$, which yields

$$0 \to \mathbb{Z} \to \mathbb{Z}[1/p] \to \mathbb{Z}(p^\infty) \to 0.$$

Under the isomorphism $\mathrm{Ext}^1(\mathbb{Z}(p^\infty), \mathbb{Z}) \cong \lim\leftarrow_n \mathbb{Z}/p^n\mathbb{Z} = \mathbb{Z}_p$, the class of this extension maps to the compatible sequence obtained by pulling back further along $\mathbb{Z}/p^n\mathbb{Z} \hookrightarrow \mathbb{Z}(p^\infty)$. The pullback of $0 \to \mathbb{Z} \to \mathbb{Z}[1/p] \to \mathbb{Z}(p^\infty) \to 0$ along $\mathbb{Z}/p^n\mathbb{Z} \hookrightarrow \mathbb{Z}(p^\infty)$ is

$$0 \to \mathbb{Z} \xrightarrow{\times p^n} \mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z} \to 0$$

where the first map is multiplication by p^n. Under the standard identification $\mathrm{Ext}^1(\mathbb{Z}/p^n\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}/p^n\mathbb{Z}$ (via the Yoneda correspondence with the projective resolution $0 \to \mathbb{Z} \xrightarrow{\times p^n} \mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z} \to 0$), Johns Hopkins University this extension corresponds to **1 mod p^n**. Ncf The resulting compatible sequence is (1 mod p, 1 mod p^2, 1 mod p^3, …), which is precisely the element $\mathbf{1} \in \mathbb{Z}_p$.

Since each p-component is 1, the full extension class is $(\mathbf{1}, \mathbf{1}, \mathbf{1}, \ldots) \in \prod_p \mathbb{Z}_p = \hat{\mathbb{Z}}$ — the image of the integer 1 under the canonical profinite completion map $\mathbb{Z} \to \hat{\mathbb{Z}}$. Wikipedia This element generates a dense subgroup of $\hat{\mathbb{Z}}$ in the profinite topology Osu but does not generate $\hat{\mathbb{Z}}$ as a $\mathbb{Z}$-module (since $\hat{\mathbb{Z}}$ is uncountable). ■

Question 2: single-prime extensions and the product decomposition

Each prime contributes the unit $\mathbf{1} \in \mathbb{Z}_p$

(a) Exactness and non-splitting. The sequence $0 \to \mathbb{Z} \xrightarrow{\iota} \mathbb{Z}[1/p] \xrightarrow{\pi} \mathbb{Z}(p^\infty) \to 0$, where ι is the inclusion and π the quotient map, is exact by construction. It does not split: a splitting would require a section $s: \mathbb{Z}(p^\infty) \to \mathbb{Z}[1/p]$, but $\mathrm{Hom}(\mathbb{Z}(p^\infty), \mathbb{Z}[1/p]) = 0$ since $\mathbb{Z}(p^\infty)$ is torsion and $\mathbb{Z}[1/p]$ is torsion-free. Alternatively, if the sequence split, then $\mathbb{Z}[1/p] \cong \mathbb{Z} \oplus \mathbb{Z}(p^\infty)$ would contain torsion, contradicting the torsion-freeness of $\mathbb{Z}[1/p] \subset \mathbb{Q}$. University of Chicago

(b) Extension class is $\mathbf{1} \in \mathbb{Z}_p$. As computed in Question 1, the pullback-to-finite-levels argument shows this extension corresponds to the compatible sequence (1, 1, 1, …) in $\lim\leftarrow_n \mathbb{Z}/p^n\mathbb{Z} = \mathbb{Z}_p$, which is the element $\mathbf{1} \in \mathbb{Z}_p$.

(c) Twisted extensions. For any $\alpha \in \mathbb{Z}_p$, the corresponding extension is the pullback of $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ along the endomorphism "multiplication by α" on $\mathbb{Z}(p^\infty)$. When $\alpha = 0$, the extension splits. When $\boldsymbol{\alpha} \in \mathbb{Z}_p^\times$ (a p-adic unit), the extension is non-split with a torsion-free middle term of rank 1. When $\alpha \in p\mathbb{Z}_p \setminus \{0\}$, the middle term has a more complex structure. The canonical extension ($\alpha = 1$) is the simplest non-split case.

The full extension as a product of prime extensions

The relationship is made precise through pullback and the Ext product decomposition. For each prime p, the diagram

$$\begin{array}{ccccccccc}
0 & \to & \mathbb{Z} & \to & \mathbb{Z}[1/p] & \to & \mathbb{Z}(p^\infty) & \to & 0 \\
 & & \| & & \downarrow & & \downarrow \iota_p & & \\
0 & \to & \mathbb{Z} & \to & \mathbb{Q} & \to & \mathbb{Q}/\mathbb{Z} & \to & 0
\end{array}$$

is a pullback square. Under the canonical isomorphism $\mathrm{Ext}^1(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) \cong \prod_p \mathrm{Ext}^1(\mathbb{Z}(p\infty), \mathbb{Z})$, the full extension maps to the tuple of its p-pullbacks: $\mathbf{(1, 1, 1, \ldots)} \mapsto \mathbf{(1_2, 1_3, 1_5, 1_7, \ldots)}$.

Conversely, one can recover the full extension from the prime extensions. Form the direct sum of extensions $\bigoplus_p E_p: 0 \to \bigoplus_p \mathbb{Z} \to \bigoplus_p \mathbb{Z}[1/p] \to \bigoplus_p \mathbb{Z}(p\infty) \to 0$, then take the **pushout along the fold map** $\nabla: \bigoplus_p \mathbb{Z} \to \mathbb{Z}$ (which sums all copies). The resulting pushout is $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$. This works because $\mathbb{Q}$ is the pushout of the diagram $\bigoplus_p \mathbb{Z}[1/p] \leftarrow^{\{\bigoplus \iota_p\}} \bigoplus_p \mathbb{Z} \to^{\nabla} \mathbb{Z}$, which identifies the diagonally-embedded copies of $\mathbb{Z}$ in each $\mathbb{Z}[1/p]$. Every rational number can be written as a finite sum of fractions with prime-power denominators, confirming that this pushout is $\mathbb{Q}$.

The universal property of the $\mathbb{Q}$-extension

The extension $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ is **universal among extensions of torsion groups by** $\mathbb{Z}$ in the following precise sense.

Theorem. *For any torsion abelian group T and any extension $0 \to \mathbb{Z} \to E \to T \to 0$, there exists a morphism of extensions*

```
0 → Z → E → T → 0
  ‖   ↓   ↓ f
0 → Z → Q → Q/Z → 0
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In particular, every such extension arises as a pullback of $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ along some homomorphism $f: T \to \mathbb{Q}/\mathbb{Z}$.

Proof. Since $\mathbb{Q}$ is injective, (Wikipedia) the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ extends to a map $\varphi: E \to \mathbb{Q}$ making the left square commute. (Wikipedia) The induced map on quotients $f: T \to \mathbb{Q}/\mathbb{Z}$ gives the desired morphism of extensions. ■

A stronger statement follows from the long exact sequence in Ext. Applying $\mathrm{Ext}(T, -)$ to $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ and using $\mathrm{Ext}^1(T, \mathbb{Q}) = 0$ ($\mathbb{Q}$ is injective) and $\mathrm{Hom}(T, \mathbb{Q}) = 0$ (T is torsion, $\mathbb{Q}$ is torsion-free), one obtains the isomorphism

$$\boldsymbol{\delta: \mathrm{Hom}(T, \mathbb{Q}/\mathbb{Z}) \to^{\cong} \mathrm{Ext}^1(T, \mathbb{Z})}$$

for every torsion group T. Every extension class in $\mathrm{Ext}^1(T, \mathbb{Z})$ corresponds uniquely to a homomorphism $T \to \mathbb{Q}/\mathbb{Z}$ — and the extension is the pullback of $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ along this homomorphism. This is the fullest sense in which the $\mathbb{Q}$-extension is universal.

Question 3: $\mathbb{Q}$ as the injective hull of $\mathbb{Z}$

Definition and proof

Definition (Eckmann–Schopf, 1953). An *injective hull* of an R-module M is a pair (E, ι) where E is an injective

R-module and $\iota: M \hookrightarrow E$ is an essential extension (En Academic) — meaning every nonzero submodule of E has nonzero intersection with $\iota(M)$.

Theorem. *$\mathbb{Q}$ is the injective hull of $\mathbb{Z}$ in the category of abelian groups.*

Proof. Three properties must be verified.

(i) $\mathbb{Q}$ is injective. Over a PID, injective modules are exactly the divisible modules (Wikipedia) (Baer, 1940). (Wikipedia) For any nonzero $n \in \mathbb{Z}$ and any $q \in \mathbb{Q}$, the element q/n satisfies $n \cdot (q/n) = q$, so $\mathbb{Q}$ is divisible.

(ii) $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is essential. Let H be a nonzero subgroup of $\mathbb{Q}$, and pick $0 \neq a/b \in H$ with $a, b \in \mathbb{Z}$, $b \neq 0$, $a \neq 0$. Then $b \cdot (a/b) = a \in H \cap \mathbb{Z}$, and $a \neq 0$. So every nonzero subgroup of $\mathbb{Q}$ intersects $\mathbb{Z}$ nontrivially. This is actually a special case of a general fact: for any integral domain R with fraction field K, the inclusion $R \subset K$ is an essential extension (Lam, *Lectures on Modules and Rings*, Example 3.35). (En Academic)

(iii) $\mathbb{Q}$ is minimal. Suppose $\mathbb{Z} \subseteq E \subseteq \mathbb{Q}$ with E injective (hence divisible). For any $a/b \in \mathbb{Q}$ with $a \in \mathbb{Z}$, we have $a \in E$, and by divisibility there exists $e \in E$ with $b \cdot e = a$, forcing $e = a/b \in E$. Hence $E = \mathbb{Q}$. ■

Uniqueness: strong in this case

The injective hull is unique up to isomorphism over the base module (Wikipedia) (Purdue University) (Eckmann–Schopf, 1953; Stacks Project, Lemma 47.5.3). (En Academic) Given two injective hulls $M \hookrightarrow E$ and $M \hookrightarrow E'$, the injectivity of E' provides an extension $\psi: E \to E'$ of the inclusion, and essentiality forces ψ to be injective; symmetry then forces ψ to be an isomorphism. (Purdue University)

In general, **this isomorphism is not unique** — different extensions yield different isomorphisms, so the injective hull does not satisfy a universal property in the categorical sense (Wikipedia) (Lam, §3E; Rotman, Theorem 3.31). However, for the specific case $M = \mathbb{Z}$ and $E = \mathbb{Q}$, the isomorphism **is unique**: any automorphism φ of $\mathbb{Q}$ fixing $\mathbb{Z}$ pointwise satisfies $\varphi(a/b) = a \cdot \varphi(1/b)$, and $b \cdot \varphi(1/b) = \varphi(1) = 1$ forces $\varphi(1/b) = 1/b$, so $\varphi = \mathrm{id}$. This rigidity follows from $\mathrm{End}_{\mathbb{Z}}(\mathbb{Q}) = \mathbb{Q}$ and the fact that the only $\mathbb{Q}$-automorphism fixing 1 is the identity.

Connection to non-splitting and the extension class

Essential extensions force non-splitting by a clean general argument.

Proposition. *If $\iota: M \hookrightarrow E$ is an essential extension with $M \neq E$, then $0 \to M \to E \to E/M \to 0$ does not split.*

Proof. If the sequence split, then $E = \iota(M) \oplus N$ for some $N \cong E/M$. (University of Chicago) Since $N \neq 0$, we have $\iota(M) \cap N = 0$, contradicting essentiality. ■

Applied to $\mathbb{Z} \hookrightarrow \mathbb{Q}$, this immediately gives non-splitting. The characterization of $\mathbb{Q}$ as injective hull also explains the extension's universality: because $\mathbb{Q}$ is the *minimal* injective module containing $\mathbb{Z}$, the quotient $\mathbb{Q}/\mathbb{Z}$ is the *largest* torsion group that can appear as a quotient of an essential extension of $\mathbb{Z}$. For torsion-free abelian groups A, the injective hull is always $E(A) = \mathbb{Q} \otimes_{\mathbb{Z}} A$ (Fuchs, *Infinite Abelian Groups*), giving a functorial description.

Question 4: the connecting homomorphism δ and $\mathrm{Ext}^1(\mathbb{Q}, \mathbb{Z}) \cong \hat{\mathbb{Z}}/\mathbb{Z}$

The long exact sequence

Applying the contravariant functor $\mathrm{Hom}(-, \mathbb{Z})$ to $0 \to \mathbb{Z} \to^i \mathbb{Q} \to^\pi \mathbb{Q}/\mathbb{Z} \to 0$ (cambridge) produces:

$0 \to \mathrm{Hom}(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) \to \mathrm{Hom}(\mathbb{Q}, \mathbb{Z}) \to \mathrm{Hom}(\mathbb{Z}, \mathbb{Z}) \to^\delta \mathrm{Ext}^1(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) \to \mathrm{Ext}^1(\mathbb{Q}, \mathbb{Z}) \to \mathrm{Ext}^1(\mathbb{Z}, \mathbb{Z}) \to 0.$ (Wikipedia)

The initial terms vanish: $\mathbf{Hom(\mathbb{Q}/\mathbb{Z}, \mathbb{Z}) = 0}$ (torsion to torsion-free) and $\mathbf{Hom(\mathbb{Q}, \mathbb{Z}) = 0}$ (any $f: \mathbb{Q} \to \mathbb{Z}$ satisfies $nf(1/n) = f(1)$ for all n, forcing $f(1)$ to be divisible by every positive integer, hence $f(1) = 0$). The terminal term vanishes: $\mathbf{Ext^1(\mathbb{Z}, \mathbb{Z}) = 0}$ since $\mathbb{Z}$ is projective. (Johns Hopkins University) This leaves the exact sequence

$\mathbf{0 \to \mathbb{Z} \to^\delta \hat{\mathbb{Z}} \to Ext^1(\mathbb{Q}, \mathbb{Z}) \to 0.}$ (cambridge) (Cambridge Core)

Computing δ(id_ℤ)

The connecting homomorphism δ sends $f \in \mathrm{Hom}(\mathbb{Z}, \mathbb{Z})$ to the Yoneda class of the pushout of $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ along f. For $\mathbf{f = id_\mathbb{Z}}$, the pushout along the identity is the original extension itself. Hence

$\mathbf{\delta(id_\mathbb{Z}) = [0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0] = (1, 1, 1, \ldots) \in \prod_p \mathbb{Z}_p.}$

For $\mathbf{f = n \cdot id_\mathbb{Z}}$ (multiplication by n), the pushout modifies the inclusion $\mathbb{Z} \to \mathbb{Q}$ by precomposing with multiplication by n, yielding an extension equivalent to n times the original class under the Baer sum. Thus $\mathbf{\delta(n) = (n, n, n, \ldots) \in \hat{\mathbb{Z}}}$, and the map $\delta: \mathbb{Z} \to \hat{\mathbb{Z}}$ is precisely the **canonical profinite completion map** $n \mapsto (n, n, n, \ldots)$. This map is injective with dense image. (Wikipedia) (University of Cambridge)

$\mathrm{Ext}^1(\mathbb{Q}, \mathbb{Z}) \cong \hat{\mathbb{Z}}/\mathbb{Z}$ and its structure

The short exact sequence $0 \to \mathbb{Z} \to \hat{\mathbb{Z}} \to \mathrm{Ext}^1(\mathbb{Q}, \mathbb{Z}) \to 0$ immediately gives $\mathbf{Ext^1(\mathbb{Q}, \mathbb{Z}) \cong \hat{\mathbb{Z}}/\mathbb{Z}}$ — the profinite integers modulo the diagonally embedded ordinary integers. This group has a rich structure:

- **Torsion-free.** If $n \cdot \bar{a} = \bar{0}$ in $\hat{\mathbb{Z}}/\mathbb{Z}$, then $na \in \mathbb{Z}$ for some $a \in \hat{\mathbb{Z}}$. Writing $a = (a_p)_p$, this means $a_p = m/n$ for all p and some fixed $m \in \mathbb{Z}$. Since $m/n \in \mathbb{Z}_p$ for every prime p, we need $p \nmid (n/\gcd(m,n))$ for all p, forcing $n \mid m$, so $a = m/n \in \mathbb{Z}$ and $\bar{a} = \bar{0}$.
- **Divisible.** For $a = (a_p) \in \hat{\mathbb{Z}}$ and $n \geq 1$, choose $m \in \mathbb{Z}$ via the Chinese Remainder Theorem such that $a_p + m \equiv 0 \bmod p^{v_p(n)}$ for each $p \mid n$. Then $b = (a + m)/n \in \hat{\mathbb{Z}}$ satisfies $n\bar{b} = \bar{a}$.
- **Cardinality $2^{\aleph_0}$.** Since $|\hat{\mathbb{Z}}| = |\prod_p \mathbb{Z}_p| = 2^{\aleph_0}$ and $|\mathbb{Z}| = \aleph_0$.

A torsion-free divisible abelian group of cardinality $\mathbf{2^{\aleph_0}}$ is a $\mathbb{Q}$-vector space of dimension $2^{\aleph_0}$, hence **isomorphic to $\mathbb{R}$ as an abstract abelian group**. (cambridge) This is Wiegold's theorem (1969, *Bull. Austral. Math. Soc.* 1, 341–343). (Semantic Scholar) (Cambridge Core)

An independent computation confirms this. Applying $\mathrm{Hom}(\mathbb{Q}, -)$ to $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$, and using $\mathrm{Hom}(\mathbb{Q}, \mathbb{Z}) = 0$ and $\mathrm{Ext}^1(\mathbb{Q}, \mathbb{Q}) = 0$, one obtains $\mathrm{Ext}^1(\mathbb{Q}, \mathbb{Z}) \cong \mathrm{Hom}(\mathbb{Q}, \mathbb{Q}/\mathbb{Z})/\mathbb{Q}$ (Cambridge Core) $\cong \mathbb{A}_f/\mathbb{Q}$,

(Neverendingbooks) where $\mathbb{A}_f = \prod'_p \mathbb{Q}_p$ is the ring of finite adèles (Boardman, *Some Common Tor and Ext Groups*, Theorem 25). (Johns Hopkins University)

What the cokernel tells us

The exact sequence $0 \to \mathbb{Z} \to \hat{\mathbb{Z}} \to \hat{\mathbb{Z}}/\mathbb{Z} \to 0$ has profound content. (cambridge) The map $\delta: \mathbb{Z} \to \hat{\mathbb{Z}}$ is the profinite completion, and its **cokernel** $\hat{\mathbb{Z}}/\mathbb{Z} \cong \mathbb{R}$ measures how far the profinite integers exceed the ordinary integers. The extension $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ generates only a "thin" line $\mathbb{Z} \subset \hat{\mathbb{Z}}$ inside the vast space of all extension classes. The remaining classes in $\mathrm{Ext}^1(\mathbb{Q}/\mathbb{Z}, \mathbb{Z})$ — those not in the image of δ — correspond to extensions $0 \to \mathbb{Z} \to E \to \mathbb{Q}/\mathbb{Z} \to 0$ where E is not isomorphic to $\mathbb{Q}$. These exotic middle terms are torsion-free rank-1 groups that are not subgroups of $\mathbb{Q}$ but rather "supernatural" completions. (cambridge)

Question 5: the Pontryagin duality diagram and the dual sequence

The complete diagram with all maps

The diagram from the nLab takes the following form, where vertical arrows denote Pontryagin duality and all groups carry natural locally compact topologies (discrete for the top row, compact or discrete as appropriate for the bottom):

$$\begin{array}{ccccc}
\mathbb{Z}(p^\infty) = \mathbb{Z}[1/p]/\mathbb{Z} & \hookrightarrow & \mathbb{Q}/\mathbb{Z} & \hookrightarrow & \mathbb{R}/\mathbb{Z} = \mathbb{T} \\
\updownarrow & & \updownarrow & & \updownarrow \\
\mathbb{Z}_p & \leftarrow & \hat{\mathbb{Z}} & \leftarrow & \mathbb{Z}
\end{array}$$

Top row (left to right, inclusions of discrete groups):

1. $\mathbb{Z}(p^\infty) \hookrightarrow \mathbb{Q}/\mathbb{Z}$: the inclusion of the p-primary component. Since $\mathbb{Q}/\mathbb{Z} \cong \bigoplus_q \mathbb{Z}(q^\infty)$, the Prüfer p-group embeds as the p-summand. (jordanbell)
2. $\mathbb{Q}/\mathbb{Z} \hookrightarrow \mathbb{R}/\mathbb{Z}$: the inclusion of the torsion subgroup of the circle. Every element of $\mathbb{Q}/\mathbb{Z}$ has finite order in $\mathbb{T}$, and $\mathbb{Q}/\mathbb{Z}$ is precisely the set of all torsion elements of $\mathbb{T}$. (jordanbell +2)

Bottom row (right to left, maps of profinite/discrete groups):

1. $\mathbb{Z} \to \hat{\mathbb{Z}}$: the canonical profinite completion map, $n \mapsto (\ldots, n \bmod m, \ldots)$, which is an injective ring homomorphism with dense image. (Wikipedia) (University of Cambridge)
2. $\hat{\mathbb{Z}} \to \mathbb{Z}_p$: the canonical projection from $\hat{\mathbb{Z}} = \prod_q \mathbb{Z}_q$ onto its p-th factor, (nLab) (Wikipedia) a surjective continuous ring homomorphism.

Vertical arrows (Pontryagin duality):

- Right column: $\mathbb{Z} \leftrightarrow \mathbb{R}/\mathbb{Z}$. The pairing is $(n, t) \mapsto e^{2\pi int}$. The discrete group $\mathbb{Z}$ and the compact group $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ are Pontryagin dual. (Harvard) (Wikipedia)
- Middle column: $\hat{\mathbb{Z}} \leftrightarrow \mathbb{Q}/\mathbb{Z}$. Since $\mathbb{Q}/\mathbb{Z} = \mathrm{colim}_n\, \mathbb{Z}/n\mathbb{Z}$ (colimit over divisibility), its dual is $\lim_n (\mathbb{Z}/n\mathbb{Z})^\wedge = \lim_n \mathbb{Z}/n\mathbb{Z} = \hat{\mathbb{Z}}$. (Wikipedia) Equivalently, $(\mathbb{Q}/\mathbb{Z})^\wedge \cong (\bigoplus_p \mathbb{Z}(p^\infty))^\wedge \cong \prod_p \mathbb{Z}_p = \hat{\mathbb{Z}}$. (Mi-ras) (Wikipedia)
- Left column: $\mathbb{Z}_p \leftrightarrow \mathbb{Z}(p^\infty)$. The compact group $\mathbb{Z}_p$ and the discrete group $\mathbb{Z}(p^\infty)$ are dual (Wikipedia) via the p-adic fractional part pairing. (Wikipedia)

Commutativity

The diagram commutes in the precise sense that **Pontryagin duality sends each inclusion in the top row to the corresponding surjection in the bottom row** (read in reverse direction). Specifically:

- **The dual of $\mathbb{Q}/\mathbb{Z} \hookrightarrow \mathbb{R}/\mathbb{Z}$ is the map $\mathbb{Z} \to \hat{\mathbb{Z}}$.** A character of $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ (i.e., an integer $n \in \mathbb{Z}$) restricts to a character of $\mathbb{Q}/\mathbb{Z}$ (i.e., an element of $\hat{\mathbb{Z}}$). This restriction map is the profinite completion $\mathbb{Z} \to \hat{\mathbb{Z}}$. (Harvard)
- **The dual of $\mathbb{Z}(p^\infty) \hookrightarrow \mathbb{Q}/\mathbb{Z}$ is the projection $\hat{\mathbb{Z}} \to \mathbb{Z}_p$.** A character of $\mathbb{Q}/\mathbb{Z}$ (an element of $\hat{\mathbb{Z}}$) restricts to a character of $\mathbb{Z}(p^\infty)$ (an element of $\mathbb{Z}_p$). This restriction is projection onto the p-th component.

This is an instance of the general equivalence **ProFinAb ≃ (TorsionAb)^{op}** given by the functor $\mathrm{Hom}(-, \mathbb{Q}/\mathbb{Z})$, which restricts Pontryagin duality to the subcategories of profinite and discrete torsion abelian groups (nLab, "Pontryagin duality for torsion abelian groups").

The dual of the exact sequence

The original sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ consists of discrete abelian groups (all three are countable, hence locally compact with the discrete topology). Pontryagin duality is an exact contravariant functor on the category of locally compact abelian groups with proper morphisms (Project Euclid) (Roeder, *Pacific J. Math.* 52(2)). Applying it contravariantly yields the exact sequence

$$\mathbf{0} \to \hat{\mathbb{Z}} \to \mathbb{A}_{\mathbb{Q}/\mathbb{Q}} \to \mathbb{R}/\mathbb{Z} \to \mathbf{0}$$

where $\mathbb{A}_{\mathbb{Q}/\mathbb{Q}}$ is the Pont (Neverendingbooks)ryagin dual of $\mathbb{Q}$ (discrete). This compact connected group is the **adelic solenoid**, described explicitly as

$$\mathbb{A}_{\mathbb{Q}/\mathbb{Q}} \cong (\mathbb{R} \times \hat{\mathbb{Z}}) / \hat{\mathbb{Z}}^\Delta$$

where $\hat{\mathbb{Z}}^\Delta = \{(n, n) : n \in \mathbb{Z}\}$ is the diagonal embedding. The maps in the dual sequence are:

- $\hat{\mathbb{Z}} \to \mathbb{A}_{\mathbb{Q}/\mathbb{Q}}$: sends $\hat{a} \mapsto [(0, \hat{a})]$, embedding the profinite integers as the fiber over the identity in $\mathbb{T}$.
- $\mathbb{A}_{\mathbb{Q}/\mathbb{Q}} \to \mathbb{R}/\mathbb{Z}$: sends $[(x, \hat{a})] \mapsto x \bmod \mathbb{Z}$, projecting to the circle.

Exactness is verified directly: the kernel of the projection consists of classes $[(n, \hat{a})]$ with $n \in \mathbb{Z}$, which equals classes $[(0, \hat{a} - n)]$ — precisely the image of $\hat{\mathbb{Z}}$. The dual of $\mathbb{Q}$ is not $\mathbb{Q}$ itself ($\mathbb{Q}$ with the discrete topology is **not self-dual**; the result of Keith Conrad, "The Character Group of $\mathbb{Q}$", Theorem 3.2, identifies the dual with $\mathbb{A}_{\mathbb{Q}/\mathbb{Q}}$

via the adelic character). The solenoid **is compact, connected, but not path-connected**; its path component of the identity is $\mathbb{R}$ (embedded densely), and it fibers over $\mathbb{T} = S^1$ with Cantor-set fibers $\hat{\mathbb{Z}}$ — a principal $\hat{\mathbb{Z}}$-bundle over the circle.

What this says about the original extension

The original sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ is an exact sequence of discrete groups. Its Pontryagin dual $0 \to \hat{\mathbb{Z}} \to \mathbb{A}_{\mathbb{Q}}/\mathbb{Q} \to \mathbb{T} \to 0$ is an exact sequence of compact groups. The duality reverses the roles: the "arithmetic" content (profinite completion, p-adic structure) that is hidden in the Ext group of the original sequence becomes geometrically visible in the solenoid, which is a nontrivial compact principal bundle. The non-splitting of the original sequence dualizes to the non-splitting of the solenoid bundle — the solenoid $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ is not isomorphic to $\hat{\mathbb{Z}} \times \mathbb{T}$.

Conclusion: what ties the five questions together

The five computations reveal a single coherent story. The extension $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ is the **canonical generator** of the profinite completion $\mathbb{Z} \subset \hat{\mathbb{Z}} = \mathrm{Ext}^1(\mathbb{Q}/\mathbb{Z}, \mathbb{Z})$, sitting as the element $(1, 1, 1, \ldots)$ in $\prod_p \mathbb{Z}_p$. Each prime contributes an independent factor $1 \in \mathbb{Z}_p$ via the pullback extension $0 \to \mathbb{Z} \to \mathbb{Z}[1/p] \to \mathbb{Z}(p^\infty) \to 0$, and these assemble into the full extension via pushout along the fold map. The characterization of $\mathbb{Q}$ as the injective hull of $\mathbb{Z}$ is what makes this extension universal: every extension of a torsion group by $\mathbb{Z}$ factors through it, captured algebraically by the isomorphism $\mathrm{Ext}^1(T, \mathbb{Z}) \cong \mathrm{Hom}(T, \mathbb{Q}/\mathbb{Z})$ for torsion T. The connecting homomorphism δ is the profinite completion map itself, with cokernel $\mathrm{Ext}^1(\mathbb{Q}, \mathbb{Z}) \cong \hat{\mathbb{Z}}/\mathbb{Z} \cong \mathbb{R}$ measuring the vast space of exotic rank-1 torsion-free groups beyond $\mathbb{Q}$. Under Pontryagin duality, the entire structure flips into the geometric world of the adelic solenoid — a compact, connected, non-trivial principal $\hat{\mathbb{Z}}$-bundle over the circle that encodes, in its topology, exactly the arithmetic content of the original algebraic extension.